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P5779

SEAT No. : 7727
[Total No. of Pages : 2

[6120]-101
First Year M.C.A. (Management)
IT-11 : JAVA PROGRAMMING
(Rev. 2020) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Figures to the right indicate full marks.

Q1) a) What do you mean by anonymous class? Explain with suitable example. [7]

b) Explain the use of "this Keyword". [3]

OR

c) What is polymorphism? Explain the compiletime & runtime polymorphism. [7]

d) What is a naming conventions for packages? [3]

Q2) a) Explain un-checked exceptions with the suitable example. [5]

b) Define array, explain two-dimensional array with suitable example. [5]

OR

c) Explain thread priorities with proper example. [5]

d) What is mean of thread-deadlock? Illustrate it with suitable example. [5]

Q3) a) Define array list. Write a program to accept search element from the user to find it from given array list. [6]

b) What is the mean of hashing? Explain with suitable example for insert operation in Hash map. [4]

OR

c) Write a note on TreeSet & TreeMap. [6]

d) Write a short on List interface. [4]

P.T.O.

Q4) a) Design following GUI. using AWT components.

[10]

The GUI form contains the following elements:

- Student Name: A text input field.
- City: A text input field with a dropdown arrow on the right.
- Gender: Two radio buttons labeled "Male" and "Female".
- Languages: Two checkboxes labeled "Data structure" and "Java".
- Submit: A button labeled "Submit".

OR

b) Demonstrate key listener interface with suitable example.

[10]

Q5) a) Create JDBC application for product table with fields Id, name & price perform insert operation & display inserted data retrieved from table to the user.

[10]

OR

b) Create a web page by using servlet to find factorical of a given number.

[10]



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M.C.A. (Management)

IT-12 : DATA STRUCTURE AND ALGORITHMS

(2020 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

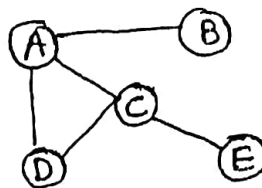
- 1) All questions are compulsory.
- 2) Figures to the right indicate full marks.

- Q1) a) Write an algorithm to reverse the nodes from singly linked list. [6]
b) Write an algorithm to copy elements from queue to stack. [4]

OR

- a) Write an algorithm to calculate sum of data of alternate nodes of doubly linked list. [6]
- b) Discuss the use of priority queue. [4]

- Q2) a) Construct binary search tree with following traversals. [6]
Preorder Traversal : 22, 15, 4, 17, 16, 19, 58, 82
Inorder Traversal : 4, 15, 16, 17, 19, 22, 58, 82
b) Write adjacency matrix and DFS for following graph. [4]
[Starting vertex : A]



OR

- a) Construct segment tree (sum of range) for following data. [6]
14, 11, 12, 16, 17, 21, 28
- b) Explain hash collision with suitable example. [4]

P.T.O.

Q3) a) Apply the rain terrace algorithm to the following problem. Input : [3, 0, 3, 0, 4, 2]. Draw the figure & find the solution. [6]

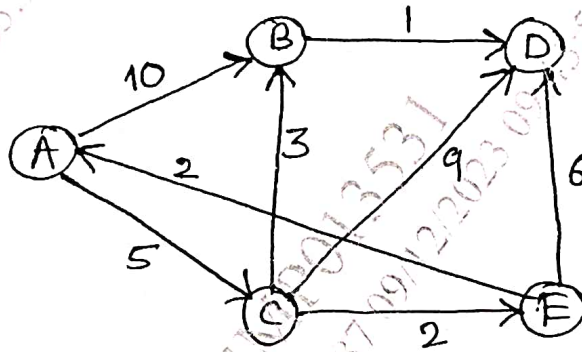
b) Describe the rules for solving N queen problem. [4]

OR

a) Apply the maximum subarray algorithm to the input : [-4, -7, -1, 4, 2, -3, 5] and find sum of maximum subarray. [6]

b) Explain combination sum problem with example. [4]

Q4) a) Apply Dijkstra's algorithm to find shortest path for following graph. [6]



b) Apply Euclidean algorithm to find GCD of 60 and 36. [4]

OR

a) Sort the following data using Mergesort algorithm [20, 55, 30, 4, 97, 13, 24]. [6]

b) Explain fast powering with suitable example. [4]

Q5) a) Find the length of longest common substring using dynamic programming for following strings. [7]

X = "congratulations" and Y = "gratitude"

b) How dynamic programming is used to find unique paths. [3]

OR

- Questi-
- a) Consider the given instance of 0/1 Knapsack problem. [7]

$n = 4, m = 8, p = (1, 2, 5, 6), w = (2, 3, 4, 5)$

Using dynamic programming determine the optimal profit and the solution vector.

- b) Explain regular expression matching using dynamic programming. [3]

□□□

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First Year M.C.A. (Management)

IT-13 : OBJECT ORIENTED SOFTWARE ENGINEERING
(2020 Pattern) (Semester-I)

Time : 2½Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All question are compulsory.
- 2) Draw neat diagrams wherever necessary.
- 3) Figures to right indicate full makrs.

- Q1) a) Explain Rapid application development model. [5]
b) Describe feature Driven development in detail. [5]

OR

- a) Explain spiral model. [5]
b) Describe web engineering process. [5]

- Q2) a) Prepare SRS as per IEEE format for G20 Summit. management system.
The overall functioning is as follows. [10]

- i) G20 member registration.
- ii) G20 member invitation.
- ii) Schedule preparation
- iv) Expenses tracking for G20 summit.

OR

- b) Prepare SRS as per IEEE format for a renowned car showroom system
for sales and service. [10]

- Q3) a) Ministry of education is organizing smart India Hackthon for developing
innovative culture in the Institute. Draw use-case diagram and class diagram
for below tasks. [10]

- i) Team registration
- ii) Selecting problem statement
- iii) Evaluating idea
- iv) Nominating the team.

OR

- b) Draw activity diagram and sequence diagram for registering a billing
complaint to MSEB. [10]

R.T.O.

- Q4)** a) Design GUI form by using appropriate UI design elements for feedback form of one day Human rights workshop conducted in your Institute. [5]
b) Draw state transition diagram for an elevator. [5]

OR

- a) Design GUI for households to request a new electricity connection. [5]
b) Draw collaboration diagram for agriculture farming system. [5]

Q5) Write short notes (any two)

[2 × 5 = 10]

- a) Inheritance and polymorphism.
c) Crystal.
d) Object diagram.

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F.Y. M.C.A. (Management)

IT-14 : OPERATING SYSTEM CONCEPTS

(2020 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) Draw the diagram whenever necessary.

Q1) a) What is CPU states? Explain different types CPU states. [5]

b) Explain physical and logical address space. [5]

OR

c) Explain Page Replacement algorithm in brief. [5]

d) What are semaphores? Explain types of semaphores. [5]

Q2) a) Explain distributed operating system. [5]

b) Explain different features of mobile operating system. [5]

OR

c) Explain No Remote Memory Access (NORMA). [5]

d) Explain installation steps in windows operating system. [5]

Q3) a) What is RTOS. Explain applications of RTOS. [5]

b) Explain Android operating system features. [5]

OR

c) Explain the features embeded operating system. [5]

P.T.O.

d) Explain below Linux command with example. [5]

i) Rmdir

ii) chmod

iii) man

iv) cat

v) ps

Q4) a) Write shell script to calculate simple interest. [5]

b) Explain the features of Linux operating system. [5]

OR

c) Write shell script to check the number is prime. [5]

d) Explain different control panel settings of windows operating systems. [5]

Q5) Write short notes on any two : [10]

a) Fragmentation

b) Process Hierarchies

c) Asymmetric Multi-processors

d) User Management in Ubuntu

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M.C.A. (Management)

IT-15 : NETWORK TECHNOLOGIES

(2020 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) All questions carry equal marks.
- 3) Draw neat diagrams wherever necessary.

Q1) a) Explain Guided transmission media in detail. [4]

b) Compare OSI model Vs TCP/IP model. [6]

OR

a) Explain Ethernet Frame Format & Access method. [4]

b) Explain Need of Layering with example. [6]

Q2) a) Draw and explain IPv6 packet format. [4]

b) Generate CRC code for the data word - 11001010 using the divisor - 10101. [6]

OR

a) For the given Class-C IP Address - 198.168.57.23 and subnet mask 255.255.255.192. Calculate [4]

i) Total no. of subnets

ii) Total no. of Host IPs/Subnet

iii) Total no. of Valid Host IPs/Subnet

iv) First and Last Valid IP for each subnet

b) Describe any two Error-detection techniques. [6]

P.T.O.

Q3) a) Find the maximum number of Hosts available on - Class-B IP address - with subnet mask 255.255.255.240. Also find maximum number of subnets available. [4]

b) Describe HDLC protocol with its sub types and frame format. [6]

OR

a) Explain performance of TCP protocol in wireless network. [4]

b) Detect and correct the single error in the received Hamming code word 110011100110 - with odd parity. [6]

Q4) a) Explain different security attacks. [4]

b) Explain MIME in detail. [6]

OR

a) Explain symmetric key cryptography with example. [4]

b) What is HTTP? Explain HTTP request and response messages in detail. [6]

Q5) a) Explain path vector protocol in brief. [4]

b) Write a TCP server socket program which accept a request from client to capitalize string and sending the response in the form of capitalized sentence back to the client. [6]

OR

a) Explain RIP in detail. [4]

b) Write Client and Server programs for establishing and termination of connection between client and server using TCP. [6]

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